

DE-GRAFT KOFI OBENG

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🌐 de-graft-kofi-obeng | 🌐 Portfolio

SUMMARY

Mathematics graduate and MSc Big Data Analytics candidate researching self-supervised graph representation learning. Undergraduate research in spectral graph theory and current work on GraphSAGE embeddings for illicit Bitcoin actor detection connect mathematical foundations with empirical machine learning. Interested in how graph structure and learning objectives influence representation quality, predictive value, and generalisation. Experience includes university teaching and data analysis.

RESEARCH INTERESTS

Graph Representation Learning • Self-Supervised Learning
Spectral and Geometric Methods for Machine Learning • Robustness and Generalisation

EDUCATION

MSc. Big Data Analytics

University of Ghana, Legon – Accra

10/2025 – Present

BSc. Mathematics

University of Ghana, Legon – Accra

01/2021 – 04/2025

Second Class Honours (Upper Division)

RESEARCH EXPERIENCE

MSc Research Project (ongoing)

Assessing the Incremental Predictive Value of Self-Supervised Graph Embeddings for Illicit Bitcoin Actor Detection: A GraphSAGE–Random Forest Framework

- Develop and evaluate self-supervised GraphSAGE representations on Elliptic++ Bitcoin graph datasets, with Random Forest for downstream classification.
- Investigate whether learned embeddings add predictive information beyond original node attributes and explicit structural features.
- Compare representation configurations and embedding dimensions using chronological evaluation and strong feature-based baselines.
- Assess predictive differences using precision–recall metrics and paired time-step bootstrap analysis.

Bachelor's Thesis: Spectral Properties of Trees

Supervisor: Dr. Kenneth Dadedzi

01/2024 – 08/2024

- Studied the eigenvalue spectra of tree graphs and their structural implications.
- Applied linear-algebraic and combinatorial techniques to characterize spectral behaviour.

Other Research Projects:

- Combinatorial Analysis of Password Possibilities — supervised by Dr. Ralph Twum.
- Application of Eigenvalues and Eigenvectors in Differential Equations — supervised by Dr. Kenneth Dadedzi.
- Applied Cryptography: Modulo Arithmetic and RSA Complexity — supervised by Dr. Ralph Twum.

SKILLS

Programming:	Python, R
Machine Learning:	Graph neural networks, self-supervised representation learning, Random Forest, model benchmarking
Data Analysis:	Data sourcing & curation, cleaning & standardization, pipeline preparation, metadata documentation, quality checks
Mathematics:	Spectral graph theory, linear algebra, combinatorics, calculus, applied cryptography
Other:	Teaching & tutoring, technical writing, research
Languages:	English (Fluent)

EMPLOYMENT

Data Analyst (Research & Curation) • <i>Pivot57</i>	<i>10/2025 – Present</i>
- Identify, catalogue, and maintain relevant data sources aligned with Pivot57’s intelligence platform. - Support data cleaning, standardization, and preparation for ingestion pipelines. - Organize datasets for dashboards and analytics; maintain structured source documentation and metadata standards. - Perform quality checks on data integrity and structure, collaborating with the technical team on ingestion.	
Graduate Teaching Assistant • <i>University of Ghana, Legon</i>	<i>09/2025 – Present</i>
Department of Mathematics	
- Courses taught: Math 121 – Algebra and Trigonometry, Math 122 – Calculus I. - Supervisor: Dr. Collins Allan.	
Teaching Assistant • <i>University of Ghana, Legon</i>	<i>10/2024 – 09/2025</i>
Department of Mathematics	
- Assisted Dr. Collins Allan in teaching Calculus I, and Algebra and Trigonometry. - Assisted Dr. Edward Koveh in teaching Calculus of Several Variables. - Organized tutorial sessions and held one-on-one sessions with students. - Marked and graded quizzes and assignments.	

AWARDS & ACHIEVEMENTS

- Valedictorian**, Class of July 2025 — University of Ghana.
- Graduated with Second Class Honours (Upper Division), BSc Mathematics.

REFERENCES

Dr. Kenneth Dadedzi Department of Mathematics University of Ghana, Legon Email: <i>(available on request)</i> Relationship: Undergraduate Thesis Supervisor	Dr. Collins Allan School of Engineering University of Ghana, Legon Email: <i>(available on request)</i> Relationship: Teaching Assistantship Supervisor
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